TFPT Minimal Kernel

Clean core note for downstream application papers

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Abstract

This note isolates the smallest TFPT theorem packet needed as a reusable core. The retained layer consists of the admissible reduced datum, the reconstructed primitive scaffold, the unique minimal carrier $E_3 \oplus E_2$, the hypercharge–seam inversion, the global Standard Model quotient

$$S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)_Y}{\mathbb{Z}_6},$$

one chiral family with ν^c from the positive half spinor, and the exact electromagnetic closure equation on the canonical branch. Everything else is intentionally left outside this kernel so that short downstream notes can extend a stable core without dragging the full theorem surface behind them.

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1 Reduced datum and primitive scaffold

Definition 1.1 (Admissible reduced theorem datum)**.** The admissible reduced datum is

$$\mathfrak{T}^{\mathrm{adm}}_{\min} := (\mathcal{A}, \mathcal{H}, D, J, \Gamma, \iota, \nu)$$

with the following built in conditions:

(i) $Z(\mathcal{A})$ is spectrally regular and $\operatorname{Spec} Z(\mathcal{A})$ is a smooth compact oriented manifold $\widetilde{M}$;

(ii) ι acts as a smooth involution of order two on $Z(\mathcal{A})$, commutes with J and Γ, and has fixed locus

$$\Sigma := \mathrm{Fix}(\iota);$$

(iii) in a collar neighborhood of Σ, the Dirac operator has product form

$$D = \gamma_n(\partial_n + B_\Sigma);$$

(iv) the cutoff profile is generated by a positive complete Bernstein measure

$$f(x) = \int_0^\infty e^{-tx^2}\, d\nu(t), \qquad d\nu(t) \geq 0;$$

(v) the seam collar is reflection regular, so the doubled operator and the later admissibility complex are well posed on the same datum.

Assumption 1.2 (Almost commutative factorization)**.** The reduced datum admits a factorization

$$\mathcal{H} \cong L^2(\widetilde{M}, S) \otimes \mathcal{H}_{\mathrm{int}},$$

where $S \to \widetilde{M}$ is the spinor bundle on the oriented double cover and $\mathcal{H}_{\mathrm{int}}$ is a finite internal Hilbert space.

Theorem 1.3 (Primitive scaffold reconstructed from the reduced datum)**.** *Let $\mathfrak{T}^{\mathrm{adm}}_{\mathrm{min}}$ be as in Theorem 1.1. Then the primitive kinematic scaffold is reconstructed canonically by*

$$\widetilde{M} = \operatorname{Spec} Z(\mathcal{A}), \qquad M = \widetilde{M}/\iota, \qquad \Sigma = \operatorname{Fix}(\iota), \qquad X = \widetilde{M} \setminus \Sigma,$$

together with the reference and relative operators

$$D_{\mathrm{ref}} := \frac{1}{2}(D + \iota D \iota), \qquad D_{\mathrm{rel}} := \frac{1}{2}(D - \iota D \iota).$$

Hence

$$\mathfrak{T}^{\mathrm{kin}}_{\mathrm{prim}} := (X, \widetilde{M}, M, \Sigma, \iota, D_{\mathrm{ref}}, D_{\mathrm{rel}}, f)$$

is derived rather than assumed. Under Theorem 1.2, the spinor bundle and the internal Hilbert space refer to the chosen factorization and are not additional independent inputs.

Remark. This note deliberately stops at the reconstructed primitive scaffold. The admissibility selector, geometric completion, transport sector, and observation layer belong to later TFPT layers and are left out here on purpose.

2 Minimal carrier theorem

Theorem 2.1 (Minimal carrier theorem)**.** *Let $E = E_b \oplus E_s$ be a split carrier with traceless diagonal generator*

$$Y_{b,s} = \operatorname{diag}\left(\underbrace{-\frac{1}{b}, \ldots, -\frac{1}{b}}_{b \text{ times}}, \underbrace{\frac{1}{s}, \ldots, \frac{1}{s}}_{s \text{ times}}\right).$$

If one simultaneously requires

$$\dim \Lambda^{\mathrm{even}} E = 16, \qquad \operatorname{Tr}_E Y_{b,s}^2 = \frac{5}{6},$$

then necessarily $\{b, s\} = \{3, 2\}$.

Proof. Set $g := b+s$. Since $\dim \Lambda^{\text{even}} E = 2^{g-1}$, the first condition gives $g = 5$. For the displayed diagonal generator,

$$\operatorname{Tr}_E Y_{b,s}^2 = b\left(\frac{1}{b^2}\right) + s\left(\frac{1}{s^2}\right) = \frac{1}{b} + \frac{1}{s} = \frac{b+s}{bs} = \frac{5}{bs}.$$

Imposing $\operatorname{Tr}_E Y_{b,s}^2 = 5/6$ yields $bs = 6$. Together with $b + s = 5$, this forces the unordered solution $\{b, s\} = \{3, 2\}$. □

From now on we therefore fix

$$E = E_3 \oplus E_2, \qquad P_3 = \frac{\mathbf{1} - \iota}{2}, \qquad P_2 = \frac{\mathbf{1} + \iota}{2},$$

and read hypercharge as

$$Y = \operatorname{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right) = -\frac{1}{3}P_3 + \frac{1}{2}P_2 = \frac{1}{12}\mathbf{1} + \frac{5}{12}\iota. \tag{2.1}$$

Lemma 2.2 (Hypercharge–seam inversion)**.** *On the minimal carrier,*

$$\iota = \frac{12Y - \mathbf{1}}{5}, \tag{2.2}$$

and therefore

$$6Y^2 - Y - \mathbf{1} = 0. \tag{2.3}$$

Consequently,

$$P_2 = \frac{\mathbf{1} + \iota}{2} = \frac{2(3Y + \mathbf{1})}{5}, \qquad P_3 = \frac{\mathbf{1} - \iota}{2} = \frac{3(\mathbf{1} - 2Y)}{5}.$$

Proof. Invert the affine relation in Equation (2.1). Squaring Equation (2.2) and using $\iota^2 = \mathbf{1}$ gives Equation (2.3). The projector formulas follow from $P_2 = (\mathbf{1} + \iota)/2$ and $P_3 = (\mathbf{1} - \iota)/2$. □

Corollary 2.3 (Carrier invariants)**.** *On the minimal branch,*

$$\gamma := \operatorname{Tr}_E Y^2 = \frac{5}{6}, \qquad B := \frac{\operatorname{rank} E_3}{\operatorname{rank} E_2} = \frac{3}{2}, \qquad B\gamma = \frac{5}{4}.$$

Remark. Because of Equation (2.3), every analytic function of Y collapses to an affine combination of $\mathbf{1}$ and Y. In that precise sense, the hard carrier is a two point algebra supported on the spectral values $-1/3$ and $1/2$.

3 Global quotient and one family

Theorem 3.1 (Global Standard Model theorem)**.** *The determinant preserving connected stabilizer of the minimal carrier split is*

$$G_{\text{SM}}^{\text{glob}} = S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)_Y}{\mathbb{Z}_6}.$$

Proof. Any automorphism preserving the split $E_3 \oplus E_2$ acts by a pair $(u_3, u_2) \in U(3) \times U(2)$. Determinant preservation on the full carrier imposes $\det(u_3)\det(u_2) = 1$, hence the connected stabilizer is exactly $S(U(3) \times U(2))$.

Now define

$$\Phi : SU(3) \times SU(2) \times U(1)_Y \longrightarrow S(U(3) \times U(2)), \qquad \Phi(g_3, g_2, z) := (z^2 g_3, z^{-3} g_2).$$

The determinant condition is automatic since $z^6 z^{-6} = 1$. Surjectivity follows by choosing $z \in U(1)$ with $z^6 = \det(u_3) = \det(u_2)^{-1}$ and setting

$$g_3 := z^{-2} u_3 \in SU(3), \qquad g_2 := z^3 u_2 \in SU(2).$$

The kernel consists of

$$\left\{ (z^{-2}\mathbf{1}_3,\ z^3\mathbf{1}_2,\ z) : z^6 = 1 \right\} \cong \mathbb{Z}_6,$$

which proves the quotient description. □

Matter is read from the positive half spinor

$$S^+ := \Lambda^{\text{even}} E, \qquad \dim S^+ = 16.$$

Theorem 3.2 (One carrier family)**.** *The positive half spinor of the minimal carrier decomposes as*

$$S^+ \cong (1,1)_0 \oplus (3,2)_{1/6} \oplus (\bar{3},1)_{-2/3} \oplus (\bar{3},1)_{1/3} \oplus (1,2)_{-1/2} \oplus (1,1)_1,$$

that is, exactly one chiral Standard Model family with ν^c. Equivalently,

$$S^+ \cong \mathbf{1} \oplus \mathbf{10} \oplus \overline{\mathbf{5}}.$$

Proof. Since

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E,$$

the sectors are read degree by degree. One has

$$\Lambda^0 E \cong (1,1)_0.$$

For the minimal carrier,

$$\Lambda^2 E \cong \Lambda^2 E_3 \oplus (E_3 \otimes E_2) \oplus \Lambda^2 E_2 \cong (\bar{3},1)_{-2/3} \oplus (3,2)_{1/6} \oplus (1,1)_1.$$

Because the full determinant is trivial in $S(U(3) \times U(2))$, one has $\Lambda^5 E \cong 1$ and hence $\Lambda^4 E \cong E^*$, namely

$$\Lambda^4 E \cong (1,2)_{-1/2} \oplus (\bar{3},1)_{1/3}.$$

Collecting the three even degree sectors yields the stated one family packet with total multiplicity $1 + 6 + 3 + 3 + 2 + 1 = 16$. □

Exterior component	SM representation	Hypercharge	Field content	Mult.
$\Lambda^0 E$	$(1,1)_0$	0	ν^c	1
$\Lambda^2 E_3$	$(\bar{3},1)_{-2/3}$	$-2/3$	u^c	3
$E_3 \otimes E_2$	$(3,2)_{1/6}$	$1/6$	Q_L	6
$\Lambda^2 E_2$	$(1,1)_1$	1	e^c	1
$\Lambda^3 E_3 \otimes E_2$	$(1,2)_{-1/2}$	$-1/2$	L_L	2
$\Lambda^2 E_3 \otimes \Lambda^2 E_2$	$(\bar{3},1)_{1/3}$	$1/3$	d^c	3
			Total multiplicity	16

Table 1: One chiral family from $S^+ = \Lambda^{\text{even}} E$, written in left chiral notation with charge conjugate singlets.

4 Electromagnetic closure and kernel package

Proposition 4.1 (Seam normalization and base opening)**.** *If the seam loop carries one unit of relative spectral flow,*

$$\mathrm{SF}(U_\Sigma) = 1, \qquad \Delta_\Sigma = 2\pi,$$

then the seam constant is

$$c_3 := \frac{\Delta_\Sigma}{16\pi^2} = \frac{1}{8\pi}.$$

Moreover the seam even base opening is

$$\varphi_{\mathrm{base}} := \frac{1-\gamma}{\pi} = \frac{1-\mathrm{Tr}_E\, Y^2}{\pi} = \frac{1}{6\pi}.$$

Definition 4.2 (Canonical abelian coefficient)**.** For family number N_{fam} and light Higgs doublet count N_Φ, define

$$b_1(N_{\mathrm{fam}}, N_\Phi) := \frac{4}{3}N_{\mathrm{fam}} + \frac{1}{10}N_\Phi.$$

On the canonical branch,

$$N_{\mathrm{fam}} = 3, \qquad N_\Phi = 1, \qquad b_1 = \frac{41}{10}.$$

Definition 4.3 (Exact seam generating function)**.** Set

$$\delta_{\mathrm{top}} := \frac{3}{256\pi^4}, \qquad x := e^{-2\alpha}.$$

The exact seam opening is

$$\varphi_\Sigma(\alpha) := \varphi_{\mathrm{base}} + \delta_{\mathrm{top}} x(1-\delta_{\mathrm{top}} x)^{-5/4}. \tag{4.1}$$

Its expansion on the admissible branch begins as

$$\varphi_\Sigma(\alpha) = \varphi_{\mathrm{base}} + \delta_{\mathrm{top}} e^{-2\alpha} + \frac{5}{4}\delta_{\mathrm{top}}^2 e^{-4\alpha} + O(e^{-6\alpha}).$$

Theorem 4.4 (Exact electromagnetic closure)**.** *On the canonical admissible branch, the fine structure constant is the unique positive solution of*

$$\alpha^3 - 2c_3^3\alpha^2 - 8c_3^6\, b_1(N_{\mathrm{fam}}, N_\Phi)\, \ln\!\big((\varphi_\Sigma(\alpha))^{-1}\big) = 0. \tag{4.2}$$

For the canonical family branch $N_{\mathrm{fam}} = 3$ and $N_\Phi = 1$, this becomes

$$\alpha^3 - \frac{\alpha^2}{256\pi^3} - \frac{41}{327680\pi^6}\ln\!\big((\varphi_\Sigma(\alpha))^{-1}\big) = 0. \tag{4.3}$$

Corollary 4.5 (Kernel output package)**.** *The minimal electromagnetic package consists of*

$$c_3 = \frac{1}{8\pi}, \qquad \varphi_0^{\mathrm{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}, \qquad \alpha^{-1}(0) = 137.03599921615843,$$

together with the seed chain

$$C_{a\gamma\gamma} = -4c_3 = -\frac{1}{2\pi}, \qquad \beta_{\mathrm{rad}} = \frac{\varphi_0^{\mathrm{ret}}}{4\pi}, \qquad \Omega_b = (4\pi-1)\beta_{\mathrm{rad}},$$

$$\lambda_C = \sqrt{\varphi_0^{\mathrm{ret}}(1-\varphi_0^{\mathrm{ret}})}, \qquad \sin^2\theta_{13} = \varphi_0^{\mathrm{ret}} e^{-\gamma}.$$

Remark. The α line must use the exact self consistent opening $\varphi_\Sigma(\alpha)$ from Equation (4.1). The retained seed φ_0^{ret} is only the frozen bridge value used for the one seed relations in Theorem 4.5.

A Optional arithmetic compression on the canonical branch

The following identity is not needed for the minimal carrier theorem itself, but it is a compact bridge relation that is often useful in short downstream notes. Define the occupancy count

$$\Omega_{\mathrm{occ}} := 16\, N_{\mathrm{fam}}.$$

Then

$$10\, b_1 = \Omega_{\mathrm{occ}}\, \gamma + N_{\Phi}.$$

On the canonical branch,

$$\Omega_{\mathrm{occ}} = \Omega_{\mathrm{adm}} = 48, \qquad \gamma = \frac{5}{6}, \qquad N_{\Phi} = 1,$$

so the hard chain becomes

$$10\, b_1 = \Omega_{\mathrm{adm}}\gamma + 1 = 41.$$

B Minimal import block for application notes

For downstream sector papers, the imported TFPT kernel can be summarized in one line as

$$\big(E_3 \oplus E_2,\, Y,\, \iota,\, G_{\mathrm{SM}}^{\mathrm{glob}},\, S^{+},\, \gamma,\, B,\, c_3,\, \varphi_{\mathrm{base}},\, \delta_{\mathrm{top}},\, b_1,\, \varphi_{\Sigma}(\alpha)\big).$$

That block is intentionally small enough to be copied into a short application note without pulling in the full closure architecture.